RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

CURATRACK: MULTI-AGENT HEALTHCARE COORDINATION SYSTEM

A PROJECT REPORT

Submitted by

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CHENNAI, TAMIL NADU

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BONAFIDE CERTIFICATE

Certified that the project report "**CuraTrack: Multi-Agent Healthcare Coordination System**" is the bonafide work of MUKESH V (240701339) and MOHAMED FAYAS MF (240701321), who carried out the project work under my supervision

SIGNATURE OF THE FACULTY INCHARGE

Submitted for the Practical Examination held on:

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SIGNATURE OF THE INTERNAL EXAMINER

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	4
1	INTRODUCTION	5
1.1	Problem Statement and Motivation	5
1.2	Objectives and Scope	6
2	LITERATURE REVIEW	7
2.1	Digital Health Records and Interoperability	7
2.2	AI in Clinical Documentation	8
2.3	Multi-Agent Systems and Safety	9
3	PROPOSED SYSTEM	10
3.1	System Requirements	10
3.2	Architecture Diagram	11
3.3	Data Flow and Agent Orchestration	12
4	MODULES DESCRIPTION	13
4.1	Patient Profile and Document Ingestion	13
4.2	Medical Information Structuring	14
4.3	Medication and Risk Monitoring	15
4.4	Emergency Coordination and Doctor Copilot	16
4.5	Privacy, Consent and Access Control	17
5	IMPLEMENTATION AND RESULTS	18
5.1	Experimental Setup	18
5.2	Prototype Workflow and Test Scenarios	19
5.3	Evaluation Metrics and Results	20
5.4	Security and Failure-Handling Tests	21
6	DISCUSSION, LIMITATIONS AND FUTURE WORK	22
7	CONCLUSION	24
8	REFERENCES	25
A	Diagram and Screenshot Placeholders	26
B	Synthetic Data and Test Cases	27

ABSTRACT

CuraTrack is a proposed privacy-aware healthcare coordination system designed to make important patient medical information easier to organize and retrieve when it is needed. Medical history can be distributed across paper prescriptions, laboratory reports, clinic records, and different hospital systems. This fragmentation can create difficulties for people living with chronic conditions and may become especially serious during emergencies when a patient is unconscious or unable to explain their medical history.

The platform allows patients or authorized caregivers to upload medical documents and maintain a unified digital health profile containing information such as allergies, current medications, chronic conditions, previous procedures, and emergency contacts. Document-processing and AI-assisted extraction can convert relevant information into structured fields while retaining source references and verification status. Specialized agents are proposed for document ingestion, record structuring, medication monitoring, risk-signal identification, emergency coordination, and concise clinician-facing summaries.

A deterministic safety layer is included so that critical actions—such as access-control decisions, mandatory emergency workflows, and rejection of unverified prescription data—do not depend solely on a language model. The design emphasizes consent-based access, least privilege, audit logging, and secure sharing of only the information required for a particular care situation. Emergency coordination aims to notify registered nearby healthcare responders and support acknowledgement and escalation, subject to connectivity and operational availability.

This report describes the motivation, related concepts, proposed architecture, modules, implementation approach, and a reproducible evaluation plan using synthetic or public data. It distinguishes proposed capabilities from verified implementation and does not claim clinical effectiveness. CuraTrack is intended to support healthcare professionals rather than replace professional judgement or emergency services.

CHAPTER 1 – INTRODUCTION

1.1 Background. Healthcare decisions depend on timely access to accurate patient information. A patient may receive care from a general practitioner, a specialist, a diagnostic laboratory, and a hospital over several years. Each encounter can generate prescriptions, reports, diagnoses, and treatment notes. When these records remain in separate systems or on paper, the patient often becomes the only link between them.

This dependence on patients carrying and explaining their own records is fragile. A person who is unconscious after an accident may not be able to identify a medication, disclose a known allergy, or describe a previous procedure. A family member or stranger who accompanies the patient may not know the relevant details. Even in non-emergency care, repeated data collection can consume time and make continuity of care harder.

1.2 CuraTrack Overview. CuraTrack proposes a secure digital health profile that brings together patient-provided documents and selected structured information. Its focus is not simply file storage: the system aims to make critical information discoverable, traceable to its source, and accessible to authorized healthcare professionals when needed. AI agents can assist with extraction and summarization, while explicit validation and access-control rules constrain what the system is allowed to do.

1.3 Motivation. The motivating use case is a patient with a chronic condition who is admitted to an unfamiliar hospital after an accident. The care team needs reliable information about known allergies, ongoing medication, important diagnoses, and previous treatment. CuraTrack aims to reduce the effort required to locate this information while acknowledging that records may be incomplete or outdated and must not be treated as infallible.

1.4 PROBLEM STATEMENT, OBJECTIVES AND SCOPE

Problem Statement. How can a privacy-preserving healthcare coordination platform organize fragmented medical documents into a structured patient profile and provide authorized clinicians with timely, source-traceable information during routine care and emergencies, while using multi-agent AI safely and keeping critical decisions under deterministic controls and human oversight?

Objectives include providing a unified digital profile for prescriptions, reports, allergies, conditions, and relevant history; exploring AI-assisted extraction from PDF and image documents; using specialized agents for distinct tasks; generating concise, source-linked summaries; supporting an emergency workflow for registered responders; protecting medical data through consent and access controls; and evaluating the prototype using synthetic or public data.

Scope. The report focuses on system design and prototype-level demonstration. It does not claim integration with every hospital, live national health infrastructure, or real ambulance dispatch. Any connection to external providers requires approved interfaces, operational agreements, and security review. The system is a support tool and must not independently diagnose a condition, prescribe medication, or delay emergency care.

Expected Contribution. The proposed contribution is a coordinated design combining structured medical information, specialized AI agents, privacy controls, deterministic safety checks, and emergency information retrieval in one workflow. The value of the design must be assessed through implementation tests and, before clinical use, appropriate clinical, usability, and security validation.

CHAPTER 2 – LITERATURE REVIEW

2.1 Digital Health Records. Electronic health records provide a digital representation of clinical information maintained by healthcare providers. Personal health records place greater emphasis on information that individuals can access or manage themselves. Both approaches can improve availability compared with paper-only processes, but usefulness depends on data quality, interoperability, access permissions, and adoption by participating organizations.

A digital record is not automatically complete. It may omit information from a clinic that does not participate in the exchange, contain outdated medication lists, or include a scanned document whose text has not been verified. CuraTrack therefore treats record completeness and provenance as explicit design concerns rather than assuming that one profile always contains every relevant fact.

2.2 Health Information Exchange. Health information exchange aims to enable appropriate clinical information to move between systems and organizations. Standards-based exchange can reduce dependence on custom interfaces, but integration involves more than matching data fields. Organizations must agree on identity matching, terminology, consent, authentication, permitted purposes, and error handling.

For CuraTrack, interoperability is a future integration objective. The proposed data model should be compatible with established healthcare exchange approaches where feasible, while avoiding any claim of live integration until the relevant interfaces have been implemented and tested.

2.3 AI-ASSISTED CLINICAL DOCUMENTATION

Clinical documents contain both structured and unstructured information. Prescriptions and reports may arrive as PDFs, photographs, scans, or typed files. Optical character recognition can detect text in images, while natural language processing can identify entities such as medication names, dates, dosages, diagnoses, and laboratory measurements. Reliability varies with image quality, handwriting, language, abbreviations, and document layout.

Large language models can assist in organizing extracted text and producing readable summaries. However, they can omit important details, misinterpret abbreviations, or produce unsupported statements. In healthcare, a plausible-sounding but incorrect allergy or dosage can be dangerous. A safe design should preserve source text, mark uncertain fields, require confirmation where necessary, and avoid silently converting uncertain output into authoritative clinical data.

Implications for CuraTrack. The system should retain the original uploaded document, store the extracted value alongside its source and extraction status, and distinguish between machine-extracted, patient-entered, and clinician-verified information. Critical information should be displayed with clear provenance. Summaries should point back to source records so clinicians can verify details.

2.4 Voice-Based Prescription Entry. Speech recognition may reduce manual typing during clinical documentation, but it introduces risks including medication-name confusion, incorrect numbers, missed negation, and errors in frequency or duration. A voice prescription workflow should create a draft, validate required fields, show the transcript and structured prescription to the clinician, and require explicit confirmation before activating a medication schedule.

2.5 MULTI-AGENT SYSTEMS AND SAFETY

A multi-agent system divides a larger task among components with distinct responsibilities. In a healthcare coordination workflow, one agent can process documents, another can structure records, another can monitor medication events, and another can prepare a clinician-facing summary. An orchestrator controls the sequence and passes structured messages between agents. This can improve modularity and make individual components easier to test, but it also creates additional interfaces and failure modes.

Agents should have narrowly defined permissions. A summarization agent does not need permission to change a prescription; a document-processing agent does not need to disclose a full record to an external responder. Structured schemas, input validation, bounded retries, timeouts, and audit events help make the workflow more predictable.

Deterministic Safety. Safety-critical behavior should not rely entirely on probabilistic model output. A deterministic policy engine can enforce authentication, authorization, mandatory SOS routing, and rejection of unverified prescription drafts. These rules should be tested independently from the language model and should remain effective if an agent returns malformed, misleading, or adversarial content.

Research Gap. CuraTrack explores the integration of source-traceable patient information, specialized agent collaboration, consent-based emergency access, and deterministic safety enforcement in a prototype-oriented workflow. This is a design objective, not a claim that no existing product or research system offers similar functions.

CHAPTER 3 – PROPOSED SYSTEM

3.1 System Overview. CuraTrack is organized as a modular application with a patient-facing interface, a protected backend, a medical document-processing pipeline, an agent orchestration service, a structured health-data store, and notification services. The design separates information extraction from authorization and emergency escalation so that an AI model cannot grant itself access or override a deterministic safety rule.

3.2 Main Workflow. First, a patient or authorized caregiver signs in and uploads a document or enters health information. The ingestion service validates the file and records metadata. OCR or document parsing extracts text; the structuring agent proposes normalized fields with source references. Validation rules identify missing, conflicting, or low-confidence values. Approved information becomes available in the patient profile. An authorized clinician can request a source-linked summary, while an SOS event invokes the configured emergency workflow.

3.3 Design Principles. Privacy by design, source traceability, human confirmation for ambiguous clinical data, separation of AI reasoning from deterministic safety, graceful handling of missing connectivity, and synthetic-only demonstration data are central principles.

System requirements include secure authentication, record-level authorization, document provenance, structured extraction, consent management, audit logging, and reliable handling of failures. The architecture should be modular so that notification providers, document processors, and language models can be replaced without changing the authorization policy.

3.1 SYSTEM REQUIREMENTS

Functional requirements define what the system must do. CuraTrack should allow an authorized user to upload a document, view processing status, review extracted fields, correct metadata, and retrieve the profile according to permission. A clinician should be able to view permitted records and request a concise summary. A valid SOS event should initiate the configured emergency workflow and record the delivery and acknowledgement status.

Non-functional requirements include confidentiality, integrity, availability, traceability, maintainability, usability, and acceptable response time. Security controls should apply at the backend even if the frontend is bypassed. Sensitive information must not be written to ordinary logs or included in public links. The application should display when a record is old, unverified, or incomplete.

Actors include the patient, authorized caregiver, registered healthcare professional, facility administrator, and system operator. Each actor should have a defined set of permissions. Administrative access should be limited to operational needs and should not automatically imply permission to read all clinical content.

Constraints include unreliable connectivity, inconsistent document quality, absence of universal provider integration, and the need for clinical review. The prototype should be designed to demonstrate a safe workflow rather than claim that every real-world healthcare dependency is already available.

3.2 SYSTEM ARCHITECTURE DIAGRAM

Insert the overall CuraTrack architecture diagram on this page. The diagram should show the patient/caregiver interface, authentication and consent service, document ingestion, OCR/PDF parsing, specialized agents, deterministic safety engine, health-record database, audit service, emergency notification service, and clinician dashboard.

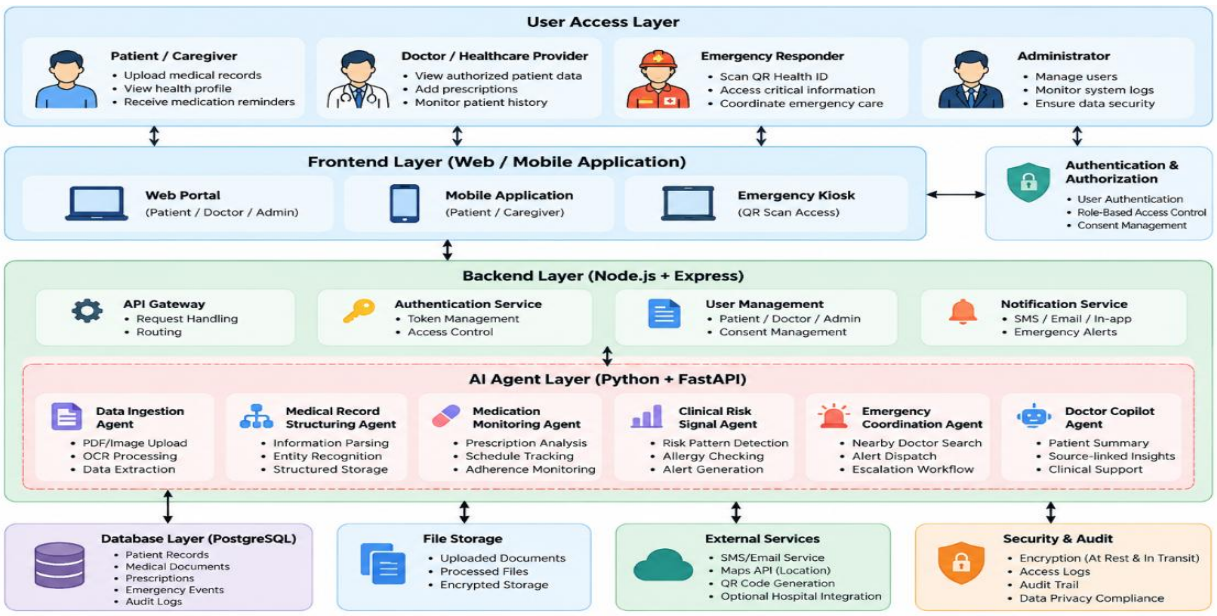


Figure 3.1: Overall System Architecture of CuraTrack

Figure 3.1. Proposed architecture of CuraTrack. Replace this placeholder with the final architecture image. Arrows should identify data movement and control flow, and the diagram should distinguish the AI-agent layer from deterministic authorization and safety controls.

Architecture layers: the presentation layer provides role-specific screens; the application layer validates requests and applies business rules; the AI layer performs bounded tasks such as document structuring and summarization; the data layer stores health information and metadata; and the security layer applies authentication, authorization, consent, encryption, and auditing across the system.

3.3 DATA FLOW AND AGENT ORCHESTRATION

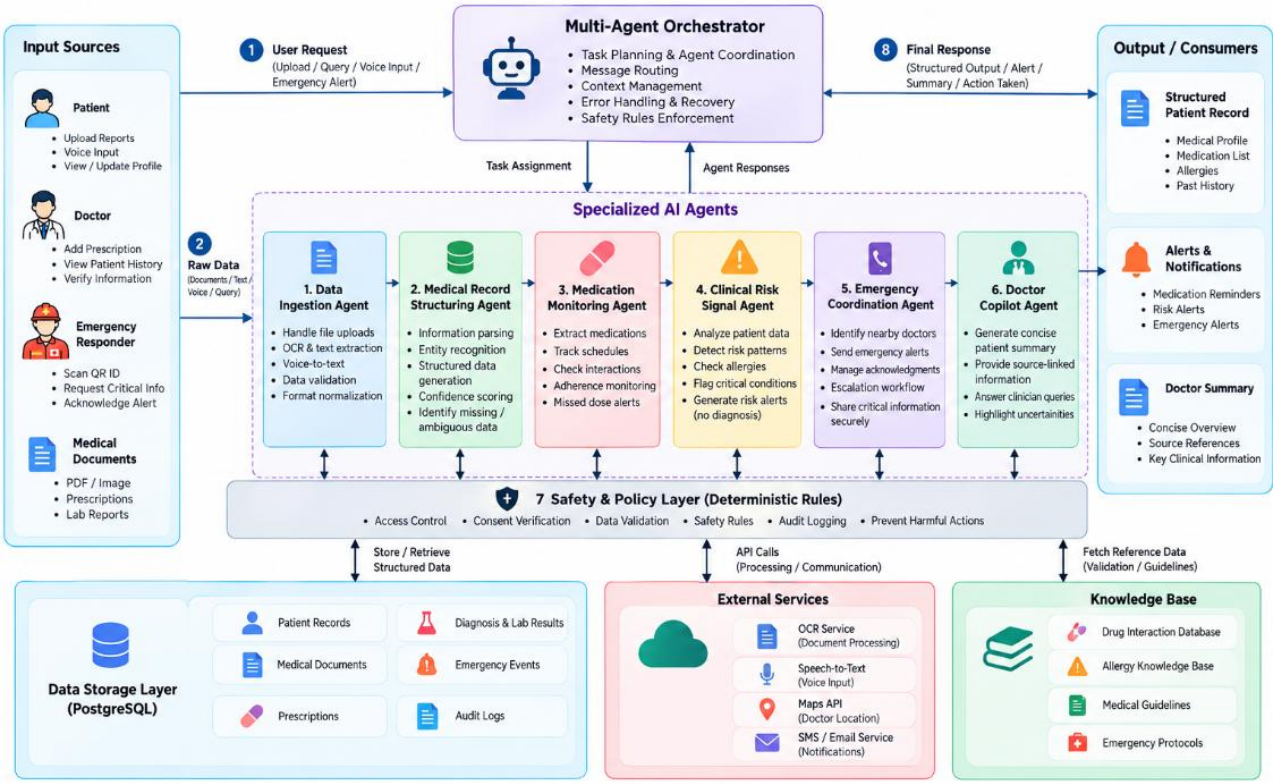


Figure 3.2: Multi-Agent Data Flow of CuraTrack

Figure 3.2. Agent collaboration and safety verification. The final image should show how ingestion, structuring, medication monitoring, risk-signal analysis, and doctor-summary outputs pass through the orchestrator and are checked against deterministic policies.

Agent Message Contract. Each agent should receive a bounded task and return a structured response containing a task identifier, status, output fields, source references, uncertainty markers where appropriate, and an error code when processing fails. The orchestrator should validate every response against a schema before using it. An agent must not be able to modify another agent's permissions or directly expose protected data.

Failure Handling. If document extraction fails, the original document remains available for manual review. If the structuring agent returns malformed data, the output is rejected and logged. If a summary agent is unavailable, the clinician can still access authorized source records. If notification delivery is not acknowledged, the workflow follows configured escalation rules rather than assuming the alert was received.

CHAPTER 4 – MODULES DESCRIPTION

4.1 Module 1: Patient Profile and Document Ingestion. The patient profile module creates a stable application identity and stores links to relevant medical documents. Patients or authorized caregivers can upload PDFs and images, add document dates, and classify files as prescriptions, laboratory reports, discharge summaries, or other records. The module validates file type, enforces upload limits, and stores documents using protected access rather than public links.

The system preserves the original file and metadata. Useful metadata includes upload time, uploader identity, claimed document date, category, and processing status. This allows the application to distinguish the date a report was created from the date it was uploaded.

Functional requirements include authentication before protected access, authorization to add or view documents, recording document metadata, showing processing failures without deleting the original file, and allowing correction of metadata subject to retention policy. The module produces a secure document record for downstream services; acceptance of a file does not prove its medical correctness.

4.2 MODULE 2: MEDICAL INFORMATION STRUCTURING

The medical information structuring module converts text extracted from documents into normalized fields. The pipeline can combine PDF text extraction, OCR for scanned pages, and an AI model for field organization. The model output is treated as candidate extraction rather than a verified medical record.

Processing stages: detect file type; extract text and page references; identify candidate entities such as medication, dosage, allergy, diagnosis, date, and laboratory result; normalize values into a defined schema while retaining original wording; validate required fields; flag contradictions; and store candidate fields with source references and verification status.

Uncertainty and Provenance. A source reference should identify the document and, where possible, the page or text span from which a value was derived. Confidence scores can help prioritize manual review, but a high model confidence score is not proof of medical correctness. When sources disagree, the interface should display the conflict and source dates rather than silently selecting one value.

All examples in this report are synthetic. The module should be evaluated using labeled test documents that contain no real patient information unless a formally approved, properly anonymized dataset is used.

4.3 MODULE 3: MEDICATION AND RISK MONITORING

The medication module organizes clinician-confirmed prescriptions into a schedule and can generate reminders. A schedule should only be activated after required prescription fields have been reviewed and confirmed by an authorized clinician. The application can record a patient's or caregiver's self-reported dose status, but it must not claim that a dose was physically taken unless a suitable verified source supports that conclusion.

Each medication record can include medicine name, strength, dosage instructions, frequency, duration, prescribing provider, start and end dates, source prescription, and verification status. Reminder events and user responses should be stored separately from the prescription itself so that a reminder is not confused with proof of adherence.

The risk-signal agent combines relevant structured data and reports patterns that may warrant professional review. For example, repeated self-reported missed doses or a concerning trend in a manually entered measurement can trigger a review notice under configured rules. Such signals are not diagnoses and should include source data, time period, and limitations.

Safety constraints: do not create confirmed medication instructions from uncertain extraction; do not independently change a prescription; do not infer that an allergy is absent because none is recorded; show the age and source of medication and allergy information; and do not delay emergency care for AI analysis.

4.4 MODULE 4: EMERGENCY COORDINATION AND DOCTOR COPILOT

The emergency module allows a patient or authorized caregiver to initiate an SOS workflow. The service captures the reported event, available location, timestamp, and minimum necessary details. A deterministic rule initiates the configured notification process. The coordination service selects registered responders based on approved criteria such as location, availability, and declared capability, where such data is available.

Responders should be able to acknowledge or decline the request. Acknowledgement must be recorded, and an unacknowledged alert must not be interpreted as a successful response. The workflow should escalate according to configured policy and should never imply that a doctor or ambulance is guaranteed to arrive.

The doctor copilot produces a concise summary of relevant, authorized records, including important allergies, current verified medications, chronic conditions, and selected previous events. Each item should link to its source and show whether it is patient-entered, machine-extracted, or clinician-verified. The clinician remains responsible for evaluating the summary against the source records and current clinical situation.

SMS should carry only a minimal alert and should not contain a full medical history. Detailed records should be accessed through an authenticated and authorized channel. Emergency access should be logged and reviewed.

4.5 MODULE 5: PRIVACY, CONSENT AND ACCESS CONTROL

CuraTrack handles highly sensitive personal information. Security must be applied across the user interface, backend, storage, AI services, and notification workflow. Role-based access control can distinguish patients, caregivers, clinicians, and administrators, but role alone is insufficient: access should also depend on the patient relationship, requested record, declared purpose, and active consent or applicable emergency policy.

Core controls include authentication, least-privilege authorization, consent grants with scope and expiry, encryption in transit and at rest, audit events for viewing and downloading records, short-lived tokens, server-side validation for QR-linked access, redaction of sensitive data from logs, and retention/deletion rules consistent with applicable law.

Deterministic rules: (1) deny access if the requester is unauthenticated or unauthorized, regardless of AI output; (2) do not activate a prescription draft with a missing or ambiguous critical field; (3) if a valid SOS event is received, invoke the configured emergency workflow independently of LLM availability; and (4) if sources conflict on an allergy or medication field, flag the conflict for review.

These rules must be implemented in application logic and tested independently. Security and safety testing should verify that AI outputs cannot override them.

CHAPTER 5 – IMPLEMENTATION AND RESULTS

5.1 Experimental Setup. The prototype can be evaluated in a controlled development environment using synthetic patient profiles, synthetic prescriptions, and generated test reports. No real patient records should be used in the demonstration. The environment should record software versions, configuration settings, test inputs, and outputs so results can be reproduced.

Candidate technology stack: React or Next.js for the interface; Node.js with Express for application APIs; PostgreSQL for structured records and audit metadata; Python with FastAPI for bounded AI tasks; a local model such as Ollama where supported; PDF parsing and OCR for documents; and an SMS provider or application notifications for alerts. Retain only technologies actually used in the final prototype and mark the rest as proposed components.

Test Environment. Record the operating system, runtime versions, database configuration, model version, document-processing library, and notification mode used for the demonstration. If a service is mocked, identify it as a mock rather than describing it as a live external integration. Credentials and real patient data must not be embedded in source code or screenshots.

5.2 PROTOTYPE WORKFLOW AND TEST SCENARIOS

The prototype workflow begins with synthetic user registration and a sample patient profile. A synthetic prescription or medical report is uploaded, processed, and converted into candidate structured fields. The interface displays extracted data with a source reference and verification status. An authorized clinician can review the record and request a concise summary.

Representative test cases include: uploading a valid synthetic PDF; processing a scanned image; extracting a sample allergy with source information; detecting conflicting medication entries; rejecting unauthorized access; producing an authorized summary; triggering SOS; handling a responder who does not acknowledge; handling a network outage; and rejecting malformed agent output.

Expected behaviour must be checked against the actual running prototype. Mark each test Pass, Fail, or Not Tested after executing it. Do not report a test as passed merely because the expected result is documented. Capture synthetic screenshots and logs that demonstrate the observed result.

Table 5.1 can be completed with the test identifier, precondition, input, expected result, observed result, and status. These records provide a reproducible basis for the implementation discussion.

5.3 EVALUATION METRICS AND RESULTS

Evaluation should measure whether CuraTrack retrieves useful information accurately, handles uncertain data safely, enforces permissions, and responds predictably when components fail. A small demonstration can establish functional feasibility, but it cannot establish clinical safety or effectiveness.

Field precision is the number of correctly extracted fields divided by all extracted fields. Field recall is the number of correctly extracted fields divided by all expected fields. The F1-score is the harmonic mean of precision and recall. Retrieval latency is measured from an authorized request to the return of the requested record. Authorization rejection rate should be calculated from controlled negative-access tests.

Additional metrics include audit completeness, schema-validation rate, successful workflow completion, notification acknowledgement rate, and recovery after network interruption. Measurements should include test-set size, test conditions, model version, and limitations.

Results template: record the measured extraction precision, recall, median retrieval latency, number of authorization tests passed out of total tests, and emergency workflow outcomes. If a metric has not been measured, report “Not measured” rather than inventing a number.

5.4 SECURITY AND FAILURE-HANDLING TESTS

Security testing should confirm that a valid login does not automatically authorize every record operation. Tests should include a patient attempting to access another patient's record, a caregiver attempting an operation outside their grant, an expired token, revoked consent, a modified record identifier, and a direct API request that bypasses the intended user interface.

Uploaded documents should be treated as untrusted input. A document may contain instructions intended to manipulate an AI model. The extraction service must process the document as data, not as authority to change system policy, reveal secrets, or grant access. Agent outputs must be schema-validated and must not contain credentials or protected internal context.

Safety tests should verify that SOS starts the configured workflow when the AI service is unavailable; excessive disclosure is blocked; incomplete prescription drafts are flagged; conflicting allergy information is displayed for review; failed notification delivery is recorded; and expired consent denies subsequent access.

For each test, record the input, expected outcome, actual outcome, logs, and remediation. Security testing must use synthetic data in an isolated environment. Any issue that could expose medical information should be treated as a release blocker until addressed.

CHAPTER 6 – DISCUSSION, LIMITATIONS AND FUTURE WORK

6.1 Discussion. A unified patient profile can reduce the need to search through separate documents when relevant history is required. Structured extraction can make important fields easier to find, while provenance allows a clinician to inspect the original source. Agent specialization can separate ingestion, structuring, monitoring, and summarization into components that can be tested independently.

The emergency workflow may help route a request to registered responders when responders are available and the communication channel works. However, the platform cannot guarantee a response and must not replace official emergency services. Its value depends on accurate registration, location quality, authorization policies, notification delivery, and a defined operational process.

6.2 Centralized Versus Federated Records. A centralized repository can simplify retrieval and provide a consistent view, but it concentrates sensitive information and creates a high-value target. A federated approach may leave records with their originating providers and retrieve them through approved interfaces, but it requires participating systems to be available and interoperable. A hybrid design may retain patient-provided copies while supporting standards-based retrieval from external providers where authorized.

6.3 Limitations. Documents may be incomplete or outdated; OCR and language models may misread medication names, dosage values, or allergies; summaries may omit context; emergency notifications depend on connectivity and responder participation; interoperability requires approved integration; and prototype testing cannot establish clinical effectiveness or regulatory compliance.

6.4 ETHICS AND FUTURE WORK

Ethical Considerations. The system should minimize collection and disclosure, explain how information is used, and provide appropriate ways to correct errors and manage consent. Access should not be granted merely because a person claims to be a healthcare professional. Emergency access needs a documented policy, appropriate authorization, audit logging, and post-event review. The system should avoid unsupported clinical recommendations and communicate uncertainty clearly.

Future development includes standards-based exchange, possible ABDM-compatible integration, improved multilingual voice input, document verification, offline synchronization, usability testing with appropriate approvals, and reliability testing under poor connectivity. Before clinical deployment, CuraTrack would require independent security assessment, clinical workflow validation, legal review, and operational agreements with participating providers.

A further research direction is to compare centralized, federated, and hybrid architectures against access latency, privacy risk, resilience, and integration effort. Another is to measure whether source-linked summaries help clinicians find critical information more efficiently without increasing the risk of overlooking original documentation.

CHAPTER 7 – CONCLUSION

CuraTrack proposes a privacy-aware approach to organizing patient medical information and making relevant details available to authorized healthcare professionals. The motivating problem is straightforward: patients should not have to rely entirely on memory or carry large collections of documents for every clinical encounter, particularly when an emergency prevents them from communicating.

The proposed design combines document ingestion, AI-assisted medical information structuring, a unified health profile, medication tracking, specialized AI agents, a doctor-facing summary, and an emergency coordination workflow. A deterministic safety layer governs authorization and selected critical actions so the system does not depend exclusively on language-model output. Provenance, uncertainty markers, consent management, and audit logs are central to the design.

The project also recognizes important limitations. Medical documents can be incomplete or misread, emergency communication can fail, and a prototype cannot demonstrate clinical effectiveness without appropriate evaluation. The system is intended to support—not replace—licensed healthcare professionals and established emergency services.

The next step is to complete the prototype, test each workflow using synthetic data, document actual results, and validate security and usability. With careful engineering and responsible integration, CuraTrack could provide a foundation for more connected, accessible, and patient-centred healthcare information management.

CHAPTER 8 – REFERENCES

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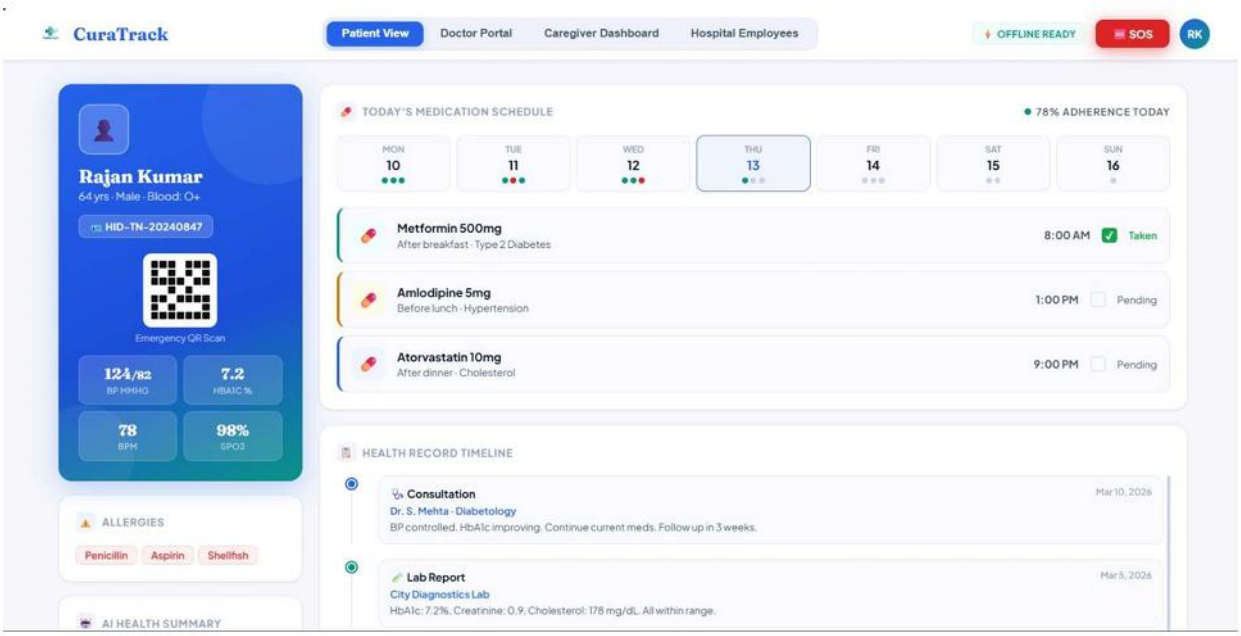
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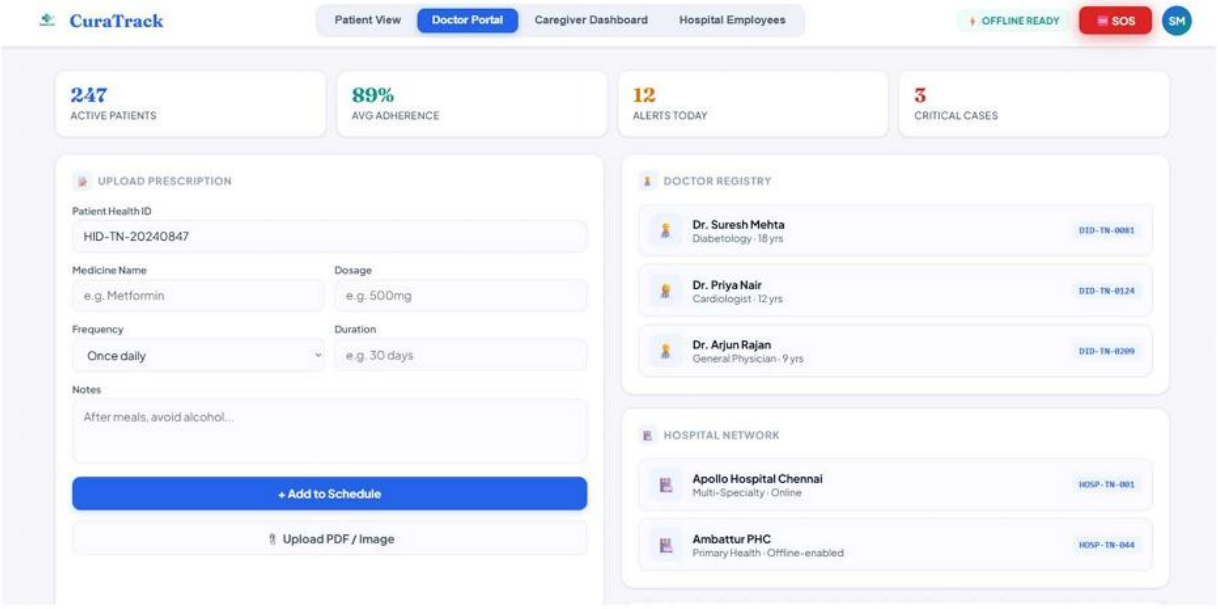
Reference note: Verify exact editions, publication details, and current legal status before submission. Add peer-reviewed papers actually consulted during the literature review; do not cite sources that were not checked.

APPENDIX A – DIAGRAMS AND SCREENSHOTS

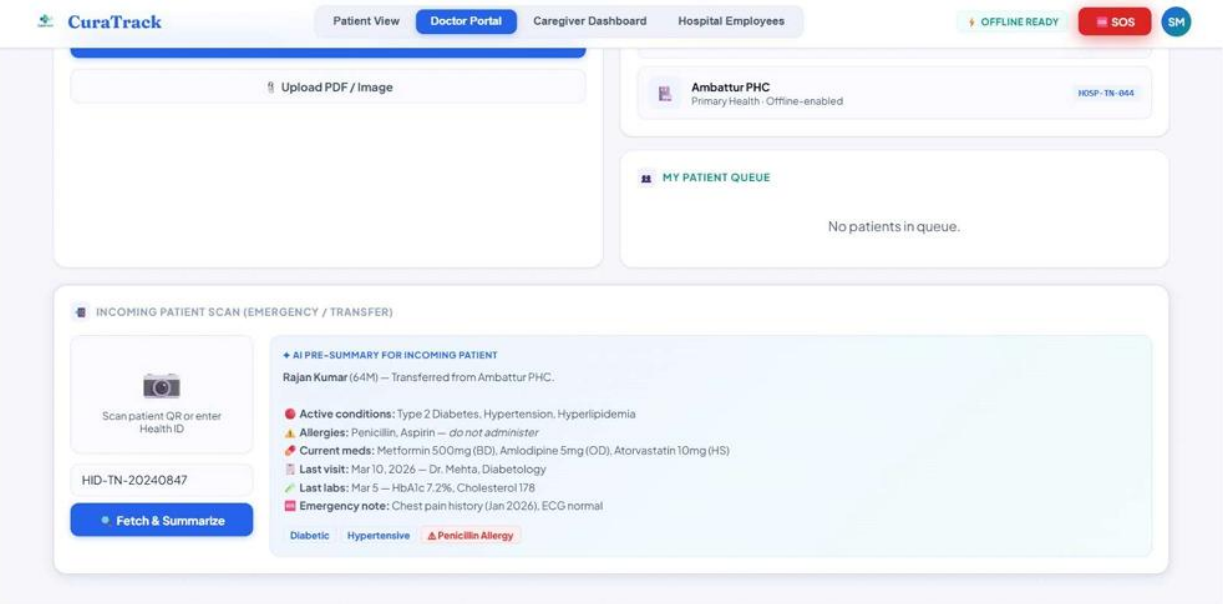
Use this page to insert the final diagrams and screenshots supplied by the project team. Each image should be numbered, have a descriptive caption, and be referenced in the main text. Ensure screenshots contain only synthetic data and do not expose real patient identifiers, access tokens, phone numbers, or credentials.



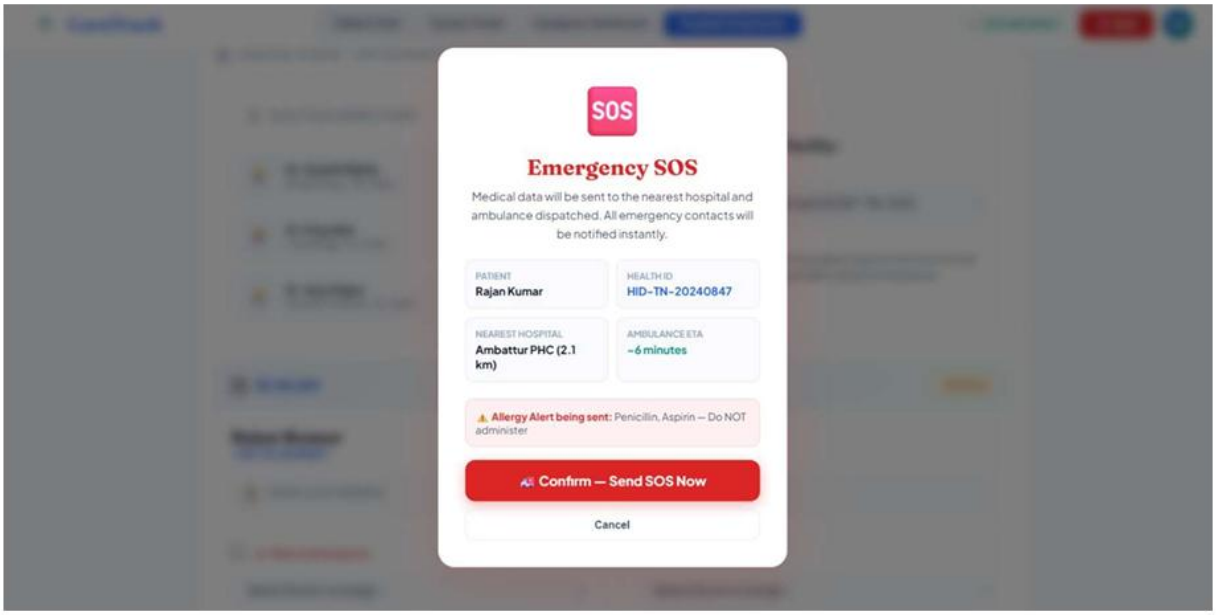
[Figure 1 : Cura Track Patient Dashboard]



[Figure 2 : Cura Track Doctor Dashboard]



[Figure 3 : Cura Track Patient Medical Report Summary]



[Figure 4 : Cura Track SOS Alert Message]

Suggested captions: CuraTrack system architecture; information flow between specialized agents and the deterministic safety layer; emergency responder notification sequence; sample patient profile and source-linked clinician summary. Replace each placeholder with the actual diagram or screenshot.

APPENDIX B – SYNTHETIC DATA AND TEST CASES

The following example is fictional and is included only to illustrate the data structure. It must not be interpreted as medical advice or a real prescription.

Example synthetic patient: CT-DEMO-001. Example fields include an allergy entry marked unverified, a synthetic medication entry requiring clinician confirmation, an example chronic condition, a synthetic source document, and a synthetic timestamp. All fields should remain clearly labelled as demonstration data.

Example Test Case: Emergency Record Retrieval. Precondition: a synthetic patient profile contains a sample allergy field, a medication entry, and source documents. An authenticated test clinician has an active authorization grant. A test emergency event is triggered.

Expected result: the application checks authorization, retrieves only permitted critical fields, displays source and verification status, creates an audit event, and provides a concise summary. If authorization is absent or expired, access is denied and the attempt is recorded. If the allergy field is unverified, it is visibly marked as unverified rather than displayed as confirmed.

Submission checklist: replace architecture and UI placeholders with actual images; confirm names and roll numbers against college records; verify technologies listed as implemented; run and record actual tests; verify references and applicable policy versions; and ask the faculty in charge to review the report before printing.